



Rossmoyne Senior High School

Semester One Examination, 2020

Question/Answer booklet

MATHEMATICS APPLICATIONS UNIT 1

Section Two:

Calculator-assumed

SOLUTIONS

WA student number: In figures

--	--	--	--	--	--	--	--

Circle your teacher's name

Buckland

Fletcher

Leonard

Pisano

Smith

Younger

Time allowed for this section

Reading time before commencing work:

ten minutes

Working time:
minutes

one hundred

Number of additional
answer booklets used
(if applicable):

--

Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet

Formula sheet (retained from Section One)

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators approved for use in this examination

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	8	8	50	52	35
Section Two: Calculator-assumed	13	13	100	98	65
Total					100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
3. You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
5. It is recommended that you do not use pencil, except in diagrams.
6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section Two: Calculator-assumed

65% (98 Marks)

This section has **thirteen** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 100 minutes.

Question 9

(6 marks)

- (a) The regular price of \$849 for a new smartphone is discounted by 15% in a sale. Determine the sale price of the smartphone. (2 marks)

Solution
$P = 849 \times (1 - 0.15)$ $= 849 \times 0.85$ $= \$721.65$ Or $P = \\$849 - \\$127.35 = \\$721.65$
Specific behaviours
✓ indicates correct multiplier or finds correct discount. ✓ correct price

- (b) After a 4% increase, the price of another smartphone was \$650. Determine the price of the smartphone before the increase. (2 marks)

Solution
$P = 650 \div (1 + 0.04)$ $= 650 \div 1.04$ $= \$625$
Specific behaviours
✓ indicates correct multiplier ✓ correct price

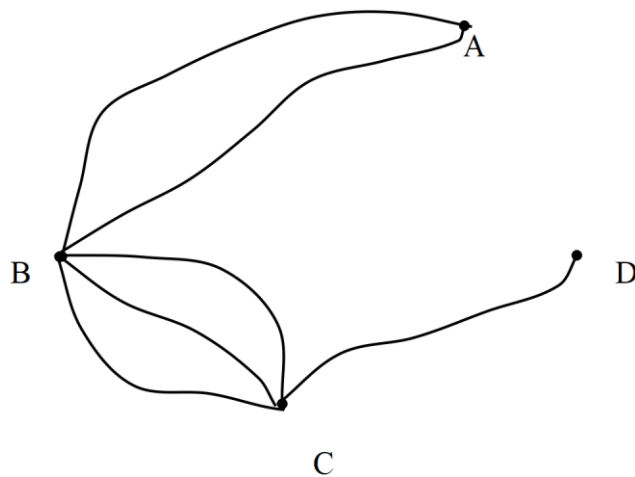
- (c) The value of a smartphone, currently \$1 200, is expected to decrease by 30% each year. Determine the expected value of the smartphone in 3 years. (2 marks)

Solution
$V = 1200 \times (1 - 0.3)^3$ $= 1200 \times 0.7^3$ $= \$411.60$
Specific behaviours
✓ indicates suitable method ✓ correct value

Question 10

(6 marks)

The network below shows roads between four towns A, B, C and D.



- a) Construct a matrix to show the number of direct links between the towns. Column and row headings should be in alphabetical order.

(2marks)

to

	A	B	C	D
A	0	2	0	0
B	2	0	3	0
C	0	3	0	1
D	0	0	1	0

- ✓ Constructs a 4x4 matrix with ordered alphabetical headings
- ✓ All elements correct

- b) Hence or otherwise construct a matrix that shows the number of ways of getting from one town to another using two roads. (4marks)

to

	A	B	C	D
A	4	0	6	0
B	0	13	0	3
C	6	0	10	0
D	0	3	0	1

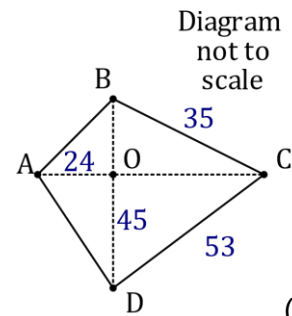
- ✓ Evidence of attempt to multiply one-stage matrix by itself.
- ✓ 6 elements correct f/t from a) if a) shows 4x4
- ✓✓ All elements correct.

Question 11

(9 marks)

The outline of a logo $ABCD$ is shown, where straight lines AC and BD intersect at right-angles at O to form quadrilateral $ABCD$.

$OA = 24$ cm, $OD = 45$ cm, $CB = 35$ cm and $CD = 53$ cm.



- (a) Determine the length of OC and the length of OB .

(3 marks)

Solution
$OC = \sqrt{53^2 - 45^2} = 28 \text{ cm}$
$OB = \sqrt{35^2 - 28^2} = 21 \text{ cm} \quad \text{No f/t}$
Specific behaviours
✓ adds lengths to existing or new diagram(s) ✓ value of OC ✓ value of OB

- (b) Determine the perimeter of logo $ABCD$.

(3 marks)

Solution
$AB = \sqrt{24^2 + 21^2} = 31.9 \text{ cm} \quad \text{f/t from student's } OB$
$AD = \sqrt{24^2 + 45^2} = 51 \text{ cm}$
$P = 31.9 + 51 + 35 + 53 = 170.9 \text{ cm}$
Specific behaviours
✓ length of AB ✓ length of AD ✓ correct perimeter

- (c) Determine the area of logo $ABCD$.

(3 marks)

Solution
$AC = 24 + 28 = 52 \quad \text{f/t from student's } OC$
$A_{ABC} = \frac{1}{2} \times 52 \times 21 = 546$
$A_{ADC} = \frac{1}{2} \times 52 \times 45 = 1\,170$
$\text{Area} = 546 + 1\,170 = 1\,716 \text{ cm}^2$
Specific behaviours
✓ clearly shows method (using 2 or 4 right-triangles) ✓ one correct part area ✓ correct total area

Question 12**(12 marks)**

A fishing store make their own conical and spherical lead weights for fishermen. To do this, they buy a block of lead costing \$335 from their supplier in the form of rectangular prism that measures $65 \times 65 \times 124$ mm.

The store then melt the lead and use casts to produce their own weights. The conical weights have a radius of 7 mm and a perpendicular height of 16 mm, and the spherical weights have a diameter of 12 mm.

The store uses the block of lead to make a batch of 250 conical and 350 spherical weights which are then sold for \$1.55 and \$1.75 respectively.

- (a) Determine the volume of lead required by the store to make all the conical weights.

(3 marks)

Solution
$V_c = \frac{\pi \times 7^2 \times 16}{3}$ $= 821.0$ $250 \times V_c = 205\,251 \text{ mm}^3$
Specific behaviours
✓ correctly substitutes ✓ volume of one cone ✓ total volume of cones <i>See note on rounding in (c)</i>

- (b) Determine the volume of lead required to make all the spherical weights.

(3 marks)

Solution
$r = 12 \div 2 = 6$ $V_s = \frac{4}{3} \times \pi \times 6^3$ $= 904.78$ $350 \times V_s = 316\,673 \text{ mm}^3$
Specific behaviours
✓ uses correct radius and substitutes ✓ volume of one sphere ✓ total volume of spheres <i>See note on rounding in (c)</i>

When a single block of lead is used to make one batch of weights, the unused lead is discarded.

- (c) Determine the volume of lead discarded when a batch is made. (2 marks)

Solution
$V_B = 65 \times 65 \times 124$ $= 523\,900$
$\text{Vol remaining} = 523\,900 - 205\,251 - 316\,673$ $= 1\,976 \text{ mm}^3$
Accept 523 900 – sum of answers in a) and b).
Specific behaviours
✓ volume of original block ✓ correct volume remaining <i>Note: Answers to (a), (b) and (c) may vary depending on level of rounding used throughout question.</i>

- (d) Given that other manufacturing costs amount to \$295 per batch, calculate the percentage profit expected by the store from selling all the weights made in a batch. (4 marks)

Solution
$\text{Income} = 250 \times 1.55 + 350 \times 1.75$ $= \$1000$
$\text{Costs} = 335 + 295$ $= \$630$
$\text{Profit} = 1\,000 - 630$ $= \$370$
$\% \text{ profit} = \frac{370}{630} \times 100 \quad \text{f/t from above}$ $= 58.7\% \approx 59\%$
Specific behaviours
✓ total income ✓ total costs ✓ profit ✓ percentage profit

Question 13

(6 marks)

- (a) A payday lender offers simple interest loans for periods of up to 90 days at a daily interest rate of 0.0685%. Determine the repayment when a loan of \$420 is taken out and repaid in full 30 days later. (2 marks)

Solution
$I = \frac{420 \times 0.0685 \times 30}{100} = \8.63
Repayment: $420 + 8.63 = \$428.63$
Specific behaviours
✓ calculates interest ✓ correct repayment

- (b) Another lender offers compound interest loans of amounts up to \$10 000 at an annual rate of 16.8%. Determine the repayment when a loan of \$3 500 is taken out and repaid in full 3 years later when interest is compounded

- (i) annually.

(2 marks)

Solution
$F = 3500(1 + 0.168)^3$ $= \$5\,576.95$
Specific behaviours
✓ indicates correct expression ✓ correct repayment

- (ii) monthly.

(2 marks)

Solution
$F = 3500 \left(1 + \frac{0.168}{12} \right)^{3 \times 12}$ $= \$5\,773.44$
Specific behaviours
✓ indicates correct expression ✓ correct repayment

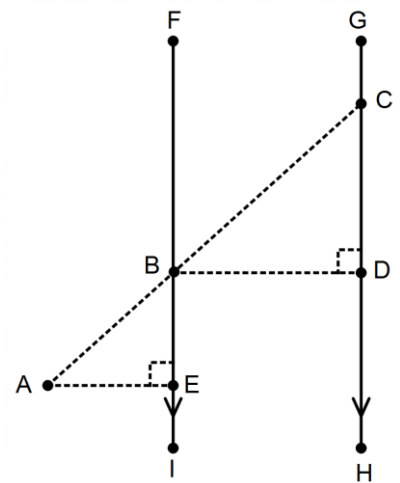
Question 14

(8 marks)

(a) For the following diagram:

FI and GH are parallel lines.

Show that the small triangle is similar to the large triangle.
Use correct labelling.



$$\angle E = \angle D \quad \checkmark \quad (= 90 \text{ degrees})$$

$$\angle ABE = \angle BCD \quad \checkmark \quad \checkmark \quad (\text{Accept } B = C \quad \text{Corresponding angles})$$

$$\therefore \triangle ABE \sim \triangle BCD \quad \checkmark \quad \text{Angle, angle condition} \quad \checkmark$$

Final statement must have corresponding vertices.

(5 marks)

(b)

If CD measures 68.5cm,

BE measures 13.7cm, and

AE measures 7.25cm

determine the measurement of BD.

$$\text{Scale ratio is } 68.5 / 13.7 = 5. \quad \checkmark \quad \text{Correct ratio. Accept } 1:5$$

$$\text{Therefore } BD = 5 \times 7.25 \text{ cm} \quad \checkmark \quad \text{Identifies } 7.25 \text{ cm}$$

$$= 36.25 \text{ cm} \quad \checkmark \quad \text{Correct distance}$$

(3marks)

Question 15**(7 marks)**

The number of people attending a show over a weekend is shown in the table below.

	Children	Adults	Seniors
Saturday	155	243	89
Sunday	212	376	175

- (a) Write the information from the table as a 2×3 matrix A .

(1 mark)

Solution
$A = \begin{bmatrix} 155 & 243 & 89 \\ 212 & 376 & 175 \end{bmatrix}$
Specific behaviours
✓ correct matrix

Admission prices for children, adults and seniors are \$22, \$48 and \$18 respectively.

- (b) Write this information as column matrix B , so that the matrix product AB will make sense.

(2 marks)

Solution
$B = \begin{bmatrix} 22 \\ 48 \\ 18 \end{bmatrix}$
Specific behaviours
✓✓ correct column matrix

- (c) Using your calculator, or otherwise, determine matrix C , where $C = AB$.

(2 marks)

Solution
$C = \begin{bmatrix} 16\ 676 \\ 25\ 862 \end{bmatrix}$
Specific behaviours
✓ 2×1 matrix
✓ correct matrix

- (d) Explain what information matrix C shows.

(2 marks)

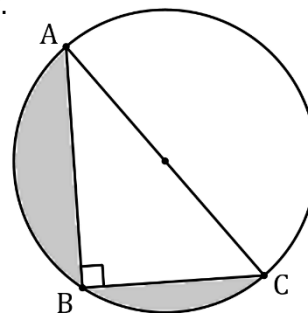
Solution
$C_{1,1}$ shows total admission fees for Saturday and $C_{2,1}$ shows total admission fees for Sunday.
Specific behaviours
✓✓ indicates total admission fees for each day

Question 16

(8 marks)

A right-triangle has vertices A , B and C on the circumference of a circle so that AC is a diameter of the circle. Two segments in the circle have been shaded grey.

Length $AB = 13.2$ cm and length $BC = 8.5$ cm.



- (a) Determine the diameter of the circle and hence the length of the arc AC . (4 marks)

Solution
$d^2 = 13.2^2 + 8.5^2$ $= 246.49$ $d = \sqrt{246.49}$ $= 15.7 \text{ cm}$ $\text{Arc } AC = \frac{1}{2} \times \pi \times d$ $= \frac{1}{2} \times \pi \times 15.7 \text{ f/t from student's } d$ $= 24.66 \text{ cm}$
Specific behaviours
✓ indicates use of Pythagoras' Theorem ✓ correct diameter ✓ indicates arc is half of circumference ✓ correct arc length (f/t from student's diameter)

- (b) Determine the grey shaded area. (4 marks)

Solution
$A_T = \frac{1}{2} \times 13.2 \times 8.5$ $= 56.1$ $r = 15.7 \div 2 = 7.85$ $A_{SC} = \frac{1}{2} \times \pi \times 7.85^2$ $= 96.8$ $A_{SHADED} = 96.8 - 56.1$ $= 40.7 \text{ cm}^2$
Specific behaviours
✓ area of triangle ✓ calculates radius (f/t from student's diameter) ✓ area of semicircle ✓ shaded area

Question 17**(6 marks)**

The currency of Fiji is the Fijian dollar. An Australian bank will sell customers 1.462 Fijian dollars for A\$1 and it will buy 1.621 Fijian dollars back for A\$1, where A\$ represents Australian dollars.

- (a) A customer had 500 Fijian dollars. Determine, to the nearest cent, how many Australian dollars they would receive from this bank. (2 marks)

Solution
$500 \div 1.621 = A\$308.45$
Specific behaviours
✓ uses buy rate ✓ correct amount

An Australian airport exchange kiosk advertises the same rates as the bank, but also charges customers a fee of 2.9% of the amount exchanged.

- (b) Determine the cost in Australian dollars for a traveller to buy 800 Fijian dollars at the airport kiosk. (2 marks)

Solution
Exchange: $F\$800 \div 1.462 = A\547.20
Add fee: $A\$547.20 \times 1.029 = A\563.06
Specific behaviours
✓ exchange amount ✓ amount with fee

- (c) Another traveller arrived back in Australia from Fiji with 225 Fijian dollars. Determine the fee that the kiosk would charge to buy this currency back, giving your answer in Australian dollars. (2 marks)

Solution
Exchange: $F\$225 \div 1.621 = A\138.80
Fee: $A\$138.80 \times 0.029 = A\4.03
Specific behaviours
✓ exchange amount ✓ correct fee

Question 18

(7 marks)

- (a) A map has a scale of 1: 75 000. The distance along a path between two campsites on the map is 32.5 cm.

- (i) Determine the actual distance between the campsites in kilometres. **(2 marks)**

Solution	
$32.5 \text{ cm} \xrightarrow{\times 75000} 2\,437\,500 \text{ cm}$	
$2\,437\,500 \text{ cm} \xrightarrow{\div 100} 24\,375 \text{ m} \xrightarrow{\div 1000} 24.375 \approx 24.4 \text{ km}$	
Specific behaviours	
✓ use scale correctly ✓ correct distance in km	

- (ii) Given that the hiker expects to walk at an average speed of 3.9 km per hour, when will they reach the second campsite if they leave the first at 1: 15 pm on Tuesday? **(2 marks)**

Solution	
$24.375 \div 3.9 = 6.25$	
$1:15 \text{ pm} + 6:15 = 7:30 \text{ pm on Tuesday.}$	
Specific behaviours	
✓ calculates time taken ✓ correct arrival time	

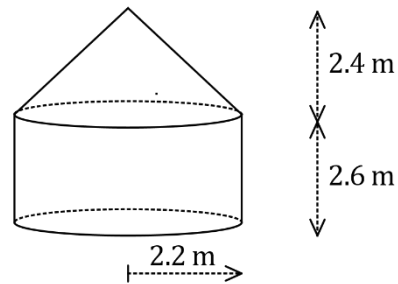
- (b) A 3D printing machine can be programmed to create copies of a solid sculpture that are mathematically similar to the original but can be any required size. The machine used 320 cm^3 of plastic to print a copy that was 4 cm tall. Determine the quantity of plastic required to print a copy that is 15 cm tall. **(3 marks)**

Solution	
Linear scale factor: $15 \div 4 = 3.75$	
Volume scale factor: $3.75^3 = 52.734375$	
Qty of plastic: $320 \times 3.75^3 = 16\,875 \text{ cm}^3.$	
Specific behaviours	
✓ indicates linear scale factor ✓ uses volume scale factor ✓ correct quantity	

Question 19**(8 marks)**

A storage silo on a farm has the form of a cone of perpendicular height 2.4 m on top of a cylinder of height 2.6 m and radius 2.2 m, as shown.

The silo stands on a level concrete floor.



- (a) Determine the volume of the silo, giving your answer to the nearest cubic metre.

(3 marks)

Solution
$V_{CYL} = \pi(2.2)^2(2.6) = 39.53$
$V_{CO} = \pi(2.2)^2(2.4) \div 3 = 12.16$
$V_{TOT} = 39.53 + 12.16$ $= 51.7 \approx 52 \text{ m}^3$
Specific behaviours
✓ volume of cylinder ✓ volume of cone ✓ correct total volume, rounded

- (b) Determine the slant height of the cone.

(2 marks)

Solution
$h = \sqrt{2.2^2 + 2.4^2}$ $= \sqrt{10.6} = 3.26 \text{ m}$
Specific behaviours
✓ indicates use of Pythagoras' Theorem ✓ correct slant height

- (c) Determine the total surface area of the outside of the silo, excluding the floor.

(3 marks)

Solution
$A_{CYL} = 2\pi(2.2)(2.6) = 35.94$
$A_{CO} = \pi(2.2)(3.26) = 22.50$
$TSA = 35.94 + 22.5$ $= 58.44 \text{ m}^2$
Specific behaviours
✓ SA of cone ✓ SA of cylinder ✓ TSA of silo

Question 20

(8 marks)

A bank offers loans of \$ P at an interest rate of 2.4% per annum to be repaid over n years.

The monthly loan repayment, M dollars, is calculated using the formula

$$M = \frac{k \times P}{500(k - 1)}, \text{ where } k = (1.002)^{12n}.$$

(a) A customer takes a loan of \$25 000 over 6 years. Determine

(i) the value of k , rounded to 4 decimal places.

(1 mark)

Solution
$k = (1.002)^{12 \times 6}$ $= 1.1547$
Specific behaviours
✓ correct value

(ii) the value of M , rounding your answer to the nearest dollar.

(2 marks)

Solution
$M = \frac{1.1547 \times 25000}{500(1.1547 - 1)}$ $= 373.2$ $\approx \$373$ (nearest dollar)
Specific behaviours
✓ indicates substitution ✓ correct value (rounding for guidance)

(b) A student has a part-time job for 52 hours a month that pays \$16.50 per hour. They have budgeted 40% of their gross monthly earnings on loan repayments and plan to take out a loan of \$9 000 for a car and repay it over 2 years. . Can the student afford this loan? Show calculations to justify your answer. (5 marks)

Solution
Loan budget is $52 \times 16.50 \times 0.40 = \343.20 per month $k = (1.002)^{12 \times 2} = 1.0491$ $M = \frac{1.0491 \times 9000}{500(1.0491 - 1)}$ $\approx \$385$ per month Advise student that they cannot afford the loan, as the repayment of \$385 is more than the \$343 budgeted each month.
Specific behaviours
✓ determines payment student can afford ✓ indicates value of k ✓ indicates substitution of values ✓ correct loan repayment ✓ advice based on calculated values

Question 21**(7 marks)**

A supermarket sells identical lollies in packs of 55, 28 and 15 for \$9.20, \$4.20 and \$2.70 respectively.

- (a) Use the unit cost method to rank the pack sizes in order from best to worst value.

(3 marks)

Solution
55-pack: $9.20 \div 55 = 0.167$ ea
28-pack: $4.20 \div 28 = 0.15$ ea
15-pack: $2.70 \div 15 = 0.18$ ea
Best to worst: 28-pack, 55-pack, 15-pack
Specific behaviours
✓ one correct unit cost ✓ all correct unit costs ✓ correct order Give this mark irrespective of method used.

- (b) The supermarket has suggested to the manufacturer that they should produce a pack containing 40 lollies.

- (i) Determine the minimum price of this pack so that it is worse value than all the existing sizes, giving your answer to the nearest 5c. **(2 marks)**

Solution
Each lolly must cost 18c:
Pack price: $40 \times 0.18 = \$7.20$, but this will be equal worst value, so must charge \$7.25 for it to be worse value.
Specific behaviours
✓ price for equal worst value ✓ correct price

- (ii) Determine the maximum price of this pack so that it is better value than all the existing sizes, giving your answer to the nearest 5c. **(2 marks)**

Solution
Each lolly must cost 15c:
Pack price: $40 \times 0.15 = \$6.00$, but this will be equal best value, so must charge \$5.95 for it to be better value.
Specific behaviours
✓ price for equal best value ✓ correct price

Supplementary page

Question number: _____

Supplementary page

Question number: _____

Supplementary page

Question number: _____

